CLAIM - Alpha as a 21|30 Closure Gate

First presentation: alpha as a closure margin from one named gate, not a fitted decimal.

This note claims that the fine-structure constant can be read as a closure margin produced by a specific 21|30 gate. The claim is conditional on a C30 carrier and a 21-step readout window; the non-trivial part is that this pair forces the rest of the ledger. Reading a 21-window on a C30 carrier forces a three-chain split, the rho=10 decagon cycle, the beat closure 210, and the first full active closure 1260.

structure

$$\alpha_{RT} = \frac{\pi}{30 \cdot 1260}(3 \cdot 30 - \Delta)$$

correction grammar

$$\Delta = AB + \frac{AB}{\rho} - \frac{1}{\rho G} - \frac{D}{\rho GCS}$$

expanded expression and bindings

$$\Delta = 2 + \frac{2}{10} - \frac{1}{10 \cdot 39} - \frac{20/21}{10 \cdot 39 \cdot 30 \cdot 7}$$

$$AB = 2, \quad \rho = 10, \quad G = 3(14-1) = 39, \quad D = \frac{20}{21}, \quad C = 30, \quad S = 7$$

numerical check / two-value status

$$\alpha_{RT} = 0.0072973525630485 \qquad \alpha_{RT}^{-1} = 137.035999201092$$

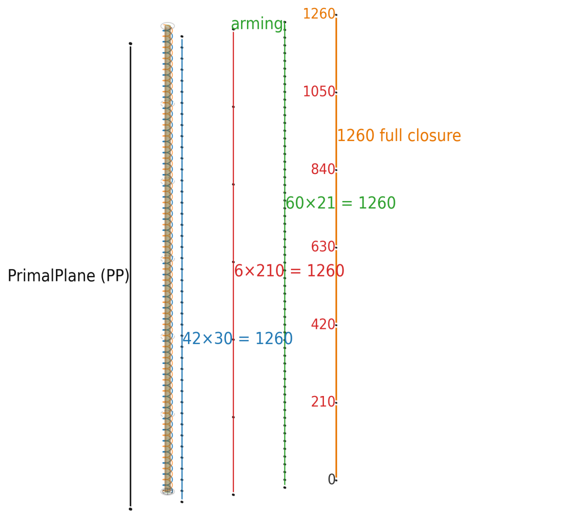
$$\Delta(\alpha^{-1}) \approx 2.41 \times 10^{-8} \qquad \alpha^{-1}_{CODATA\ 2022} = 137.035999177(21)$$

Gate value = ideal closure margin; CODATA = operational laboratory value. Offset = explicit target, not hidden rounding.

Full closure

1260 = 21 x 3 x 20 = 6 x 210 = 42 x 30

Full closure rule: first N where 3 chains carry 20 active windows each.
Thus N/21 = 3 x 20 = 60, so N = 1260.

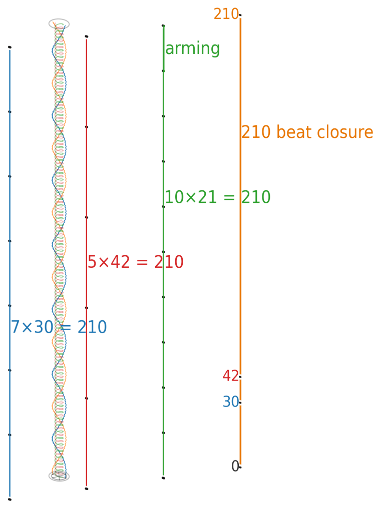


Bars and gradation ticks retained: 42 x 30, 6 x 210, 60 x 21.

Beat closure

210 = 7 x 30 = 5 x 42 = 10 x 21

210 is the local beat: lcm(21,30)=210.
10 windows: 210/21=10; 7 C30 legs close with 5 forty-two waves.



Local cell where the 21|30 gate becomes visible.

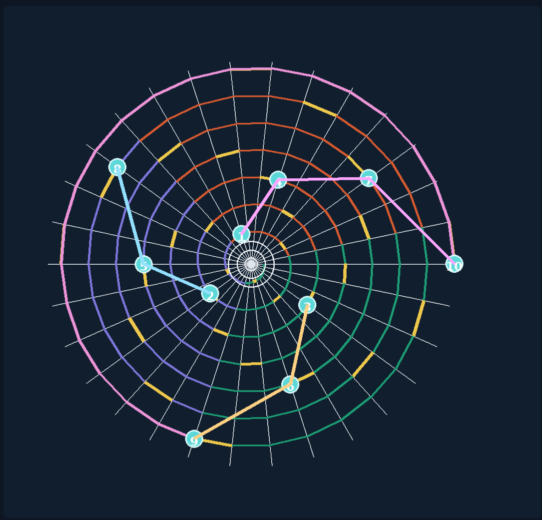
Break condition Identify a link invalid, non-unique under stated rules, or replaceable by a stricter construction.

THE 21|30 GATE - Break the Ledger

The point is not that the numbers are interesting; the point is that the links co-lock.

One beat: the 21|30 gate

Intentionally non-physical display. Geometry is unfolded for readability.
Visual aid only; the claim stands or falls on the ledger links.



full spiral visible: 210 = 10 x 21 = lcm(21,30)

W21 on C30
read a 21-window on a 30-slot carrier

gcd(21,30) = 3
forced split into three transport chains

30 / 3 = 10
rho / decagon readout, 36 degrees per chain step

lcm(21,30) = 210
one local beat closure

N = 21 x 3 x 20 = 1260
full closure rule: first N with 3 chains x 20 active positions

G = 3(14 - 1) = 39
42 = 3 x 14, then one seam excluded per chain

Companion visualizer (not part of the claim): package file RT_ALPHA_READOUT_GEOMETRY.html; GitHub path: https://github.com/Bo-Fremling/rhythm-theory-release/blob/Release/RT_ALPHA_READOUT_GEOMETRY.html. It is only an interactive reader aid; the claim stands or falls on the ledger links below.

Gate link	Consequence
3 = gcd(21,30)	forced split into three transport chains
10 = 30 / 3	rho = 10 / decagon readout
210 = lcm(21,30)	one local beat closure
1260 = 21 x 3 x 20 = 6 x 210 = 30 x 42	full closure rule: first N with 3 chains x 20 active positions
42 = 1260 / 30 = 3 x 14	C30 carrier turns under the three-chain split
G = 3 x (14 - 1) = 39	one excluded seam per transport chain

Correction grammar, not term fitting

Generated in order: A/B polarity -> rho-readout -> seam-gap subtraction -> final gate-intersection.

Delta = AB + AB/rho - 1/(rho G) - D/(rho G C S)

Allowed factors must already be forced by the 21|30 gate: AB=2, rho=10, G=39, D=20/21, C=30, S=7.

Expansion: Delta = 2 + 2/10 - 1/(10*39) - (20/21)/(10*39*30*7).
Critical link: 42=3 x 14; one seam excluded per chain gives G=3 x (14-1)=39.

A stricter competing grammar would refute or improve this claim.

Final claim

alpha_RT is proposed as a closure-geometric entity, structurally analogous to pi in the limited sense that both are dimensionless closure measures: pi measures circular closure; alpha_RT measures closure defect in the RT release ledger. To refute it, break one named link under the stated rules. Non-uniqueness test: a different primitive gate must generate the same closure value using only its own forced links and an equal-or-stricter grammar; free recombinations do not count.